

A Five-Year Review of Patients Managed in the Physiotherapy Clinic of the Geriatric Center of the University College Hospital, Ibadan

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ABSTRACT

According to the World Health Organization, the global population aged 60 years and above was approximately one billion in 2019 and is projected to increase to 1.4 billion by 2030 and 2.1 billion by 2050. This rapid demographic expansion, particularly in developing countries, underscores the growing importance of geriatric care, including physiotherapy, in maintaining functional independence and quality of life. Clinical audits, which provide objective evaluations of healthcare services, are vital tools for ensuring quality improvement in physiotherapy practice. This study is aimed to review the physiotherapy management patterns and treatment outcomes among elderly patients at the Chief Tony Anenih Geriatric Centre, University College Hospital, Ibadan. This retrospective study analyzed case files of patients managed at the Physiotherapy Department of the Centre between January 2018 and December 2022. Data obtained included demographics, referral details, diagnoses, treatment modalities, outcome measures, home programmes, session frequency, and discharge methods. Descriptive statistics such as means, percentages, and frequencies were used for data summarization. A total of 321 patient files were reviewed. Most patients were aged 60–75 years (80.7%), with females representing 74%. Delayed presentation (35.8%) and multimorbidity (27.7%) were observed. Osteoarthritis (16.8%), cervical/lumbar spondylosis (14%), and stroke (9.7%) were the most common conditions. Exercise and manual therapy were universally applied, while electrotherapy and thermotherapy were also common. There was a good use of standardized outcome measures. The study revealed presentation of health conditions commonly reported among older and conventional physiotherapy usually applied in their managements, Regular audits and improved utilization of outcome measures are recommended to enhance geriatric physiotherapy care.

Keywords: Geriatric patients, Geriatric center, Physiotherapy management

INTRODUCTION

Geriatrics is a specialized field of medicine that addresses the prevention, diagnosis, and management of health conditions in older adults, typically those aged 65 years and above.¹ In 2019, individuals within this age group constituted 9% of the global population, with projections indicating a rise to 16% by 2050.² In Nigeria, older adults accounted for 4% of the population according to the 2006 census.³ The geriatric discipline is inherently interdisciplinary, drawing on expertise from medicine, nursing, physiotherapy, occupational

therapy, pharmacy, and social work to optimize health outcomes, preserve independence, and delay institutionalization.^{4,5}

Specialized geriatric centers adopt a collaborative, team-based approach to address the multifaceted medical, psychological, and social challenges associated with aging.⁶ The Chief Tony Anenih Geriatric Centre (CTAGC), established in 2012 at the University College Hospital, Ibadan, Nigeria in sub-Saharan Africa, is a dedicated geriatric facility. The centre offers multidisciplinary services including ophthalmology, psychiatry, nutrition, dentistry, and physiotherapy. The physiotherapy unit, inaugurated in 2013, provides inpatient and outpatient rehabilitation for musculoskeletal and other age-related conditions. To date, however, no audit has been conducted to systematically evaluate patient characteristics, morbidity patterns, and therapeutic interventions in the physiotherapy

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Date manuscript was received: 29.12.2025

Date manuscript was accepted: 4.1.2026

unit. Clinical audits remain essential tools for benchmarking care, identifying service gaps, and driving quality improvement.⁷

A study reported that most common presenting morbidities at this CTAGC were mostly noncommunicable diseases such as hypertension and musculoskeletal conditions.⁸ Musculoskeletal conditions are a leading contributor to disability worldwide. The treatment of these conditions is often provided by physiotherapists.⁹ Consequently, structured audits of physiotherapy practice in geriatric settings are imperative to ensure evidence-based, patient-centered care and improved clinical outcomes.

MATERIALS AND METHODS

The following materials were used for data collection:

The physiotherapy case entry ledger used between 2018 and 2022. The following information were retrieved from the register: (a) Name of patient, (b) Hospital number, (c) Condition of patient/diagnosis. The physiotherapy treatment casefiles which had the following information:

Year (2018-2022), Age of patient, Sex of patient, Date of referral for physiotherapy, Date of first reporting at physiotherapy, Source of history and history, Diagnosis, Plan of treatment indicated, Type of modality used, Type of outcome measure used, home programmes prescription, How often reviewed, Appointment pattern, Number of treatment sessions and Discharge method.

Study Location

This study was conducted at the Physiotherapy unit of Chief Tony Anenih Geriatric Centre of the University College Hospital, Ibadan.

Research Design

This study was a retrospective survey that required retrieval of information from case files of patients at the physiotherapy unit of CTAGC at the University College Hospital, Ibadan

between 1st of January 2018 and 31st of December 2022.

Procedure for Data Collection

Ethical approval was obtained from the University of Ibadan and University College Hospital (UI/UCH) Health Research Ethics Committee before carrying out the study. Files were handled with utmost confidentiality. A spreadsheet was purposely designed to enter information obtained from the case files of patients at the physiotherapy unit of CTAGC at the University College Hospital, Ibadan. Case files of identified patients were retrieved and reviewed. The ages of the patients were grouped using the WHO classification.

Data Analysis

The information from the case files were analyzed using the Statistical Package for Social Sciences version 26 software (SPSS, IL, Chicago). Descriptive statistics of frequency, percentage and mean were used to summarize the data of the patients and their clinical information.

RESULTS

Socio-demographic Characteristics of the Patients

A total of 462 patient names and hospital numbers from the Physiotherapy Unit of the Chief Tony Anenih Geriatric Centre, University College Hospital, between January 2018 and December 2022, were retrieved from the case-entry ledger. Of these, 87 case files were untraceable and 54 had insufficient information, leaving 321 complete case files for review. The mean age of patients was 69.5 ± 7.1 years, with the 60–75 years age group being most represented (80.7%; $n = 259$). The majority were females (74%; $n = 238$), and in most cases, patients provided their own medical history (95.3%; $n = 306$). Other socio-demographic characteristics are detailed in Table 1.

Diagnoses of the Patients

The diagnoses of the patients managed during this period at the Physiotherapy unit of Chief Tony Anenih Geriatric Center are presented in Table 2. Osteoarthritis (OA) accounted for the highest number of conditions managed at the clinic with 16.8% (n = 54), while spondylosis followed with 14% (n = 45). Parkinson's disease accounted for the lowest number of conditions managed with 1.6% (n = 5). Patients with more than one condition (e.g. knee OA + lumbar spondylosis) were represented with 'multiple diagnosis' which revealed the prevalence of multi-morbidity in the older persons, while 16.5% (n = 53) of the total case notes had missing diagnosis and was represented with 'Not documented'. The list of conditions categorized under the multiple diagnoses include conditions such as Sacroiliac joint dysfunction (SIJ dysfunction), trigger finger, spinal dysfunction, postural syndrome,

adhesive capsulitis, derangement syndrome, cognitive impairment, stenosing tenosynovitis, cancer, balance and coordination impairment, myeloma, trauma from road traffic accident, scoliosis, piriformis syndrome etc

Yearly Distribution between 2018 and 2022

The yearly distribution of geriatric patients showed that 2022 recorded the highest attendance with 104 cases (32.3%), while 2018 had the lowest with 25 cases (7.8%). Male attendance peaked in 2018 (30%), whereas the highest female attendance occurred in 2021 (79%). Regarding diagnoses, lumbar spondylosis was the most common condition in 2018 (24%; n = 6), while osteoarthritis (OA) was the predominant condition from 2019 to 2022, reflecting its consistent prevalence among older adults attending the physiotherapy clinic during the study period as shown in Table 3 and 4.

Table 1 Socio-demographic Characteristics of Geriatric Patients (N = 321)

Variables	Categories	Frequency (n)	Percentage (%)
Age group of patients (years) using WHO classification	60-75	259	80.7
	76-90	60	18.7
	>90	2	0.6
Sex	Female	238	74
	Male	83	26
Source of history	Patients	306	95.3
	Case notes and	5	1.6
	Caregivers		
	Not documented	10	3.1

Table 2. Diagnosis of Geriatric Patients (N = 321)

Diagnosis	Frequency(n)	Percentage (%)
Shoulder/Hip/Knee Osteoarthritis	54	16.8
Lumbar spondylosis	45	14.0
Stroke	31	9.7
Cervical spondylosis	10	3.1
Parkinson's disease	5	1.6
Other conditions	34	10.6
Multiple Diagnosis	89	27.7
Not documented	53	16.5

Table 3. Yearly Distribution of Geriatric Patients from 2018-2022 (N = 321)

Year	Frequency(n)	Percentage (%)
2018	25	7.8
2019	78	24.3
2020	66	20.6
2021	48	15
2022	104	32.3

Management Pattern of Geriatric Patients between 2018 and 2022

The findings presented in Table 5 show that most geriatric patients (35.8%; n = 115) reported to the physiotherapy clinic between two and four months after referral, indicating a delay in seeking treatment. However, all patients commenced therapy on the same day they presented at the physiotherapy unit. The main physiotherapy modalities employed included exercise therapy and manual therapy, both of which were used in all cases. Other modalities included thermotherapy, cryotherapy, and electrotherapy, with electrotherapy being the least utilized (4.7%). Each patient's treatment plan included prescribed home exercise programmes. Exercise therapy involved aerobic, resistance, flexibility, balance, stability, and core strengthening exercises aimed at improving function, reducing pain, and enhancing mobility. Manual therapy techniques comprised joint and soft tissue mobilization, proprioceptive neuromuscular facilitation (PNF), spinal manipulation, traction, and the McKenzie technique. Thermotherapy was used in more than three-quarters of the cases, making it more common than cryotherapy, which appeared in only forty-one cases, suggesting a general preference for heat-based modalities among physiotherapists.

In terms of follow-up, discharge methods were poorly documented in 84.1% (n = 270) of cases, while 14.7% (n = 47) of patients were still receiving treatment at the time of data collection. No self-discharges or deaths were recorded. Furthermore, 77.9% (n = 250) of

patients were not reviewed after initial sessions, while only 18.1% (n = 58) were reviewed once, even after about twenty sessions. Assessment using standardized outcome measures was carried out in 77.6% (n = 249) of cases, while 22.4% (n = 72) lacked any documented outcome evaluation

DISCUSSION

This study examined the cases managed at the Physiotherapy Department of the Chief Tony Anenih Geriatric Centre (CTAGC), University College Hospital, Ibadan, between 2018 and 2022. The findings revealed that a large proportion of the patients were aged between 60 and 75 years. This age group represents the largest segment of the ageing population in Nigeria (Nigeria Data Portal, 2019). Similar findings were reported by Cadmus et al.,⁸ who found that about 71.7% of their participants were within the same age bracket. The predominance of this age group could be attributed to the fact that mortality rates rise with advancing age¹⁰ thus, individuals aged 60–75 years constitute the youngest segment within the older population and are more predominant among older persons who seek physiotherapy services than those aged 75 year and above.

A key observation from this study was the higher proportion of female geriatric patients compared to males. This aligns with global data showing that women generally have longer life expectancies than men.¹¹ A previous study at CTAGC report high prevalence of multi-morbidity, particularly hypertension, with notable gender-related differences in

disease presentation, women having significantly higher proportion.⁸

Table 4. Yearly Distribution of Gender, age-group and Condition from 2018-2022

Variables	Categories	2018	2019	2020	2021	2022
Gender						
	Female	17(68%)	61(78%)	46(70%)	38(79%)	76(73%)
	Male	8(32%)	17(22%)	20(30%)	10(21%)	28(27%)
Age group						
	60-75	21(84%)	58(74%)	59(89%)	34(71%)	86(83%)
	76-90	4(16%)	18(23%)	7(11%)	14(29%)	18(17%)
	>90	0(0%)	2(3%)	0(0%)	0(0%)	0(0%)
Conditions						
	OA	1(4%)	15(19%)	11(16.5%)	6(13%)	21(20%)
	LS	6(24%)	10(13%)	8(12%)	7(15%)	14(13%)
	Stroke	0(0%)	8(10%)	11(16.5%)	2(4%)	10(10%)
	CS	0(0%)	3(4%)	4(6%)	0(0%)	3(3%)
	PD	1(4%)	0(0%)	1(2%)	1(2%)	2(2%)
	Others	4(16%)	9(12%)	8(12%)	4(8%)	9(9%)
	Multiple	10(40%)	21(27%)	13(20%)	17(35%)	28(27%)
	ND	3(12%)	12(15%)	10(15%)	11(23%)	17(16%)

Keys: OA: (Shoulder/Hip/Knee) Osteoarthritis, LS: Lumbar Spondylosis, CS: Cervical Spondylosis, PD: Parkinson's disease, Others: Other conditions, Multiple: Multiple conditions, ND: Not Documented

Table 5: Management pattern of Geriatric patients (N = 321)

Variables	Categories	Frequency(n)	Percentage (%)
Time difference between referral and presentation at Physiotherapy	Same day	2	0.6
	1-10 days	8	2.5
	11-20 days	17	5.3
	21-30 days	37	11.5
	1-2 months	41	12.8
	2-4 months	115	35.8
	>4 months	4	1.3
	Not documented	97	30.2
Time difference between presentation and commencement of treatment	Same day	321	100
Modality used	Exercise therapy	321	100
	Manual Therapy	321	100
	Thermotherapy	261	81.3
	Cryotherapy	41	12.8
	Electrotherapy	15	4.7
Plan of treatment	Yes	321	100
	No	0	0
Home programme prescription	Yes	321	100
	No	0	0
Discharge pattern	Not documented	270	84.1
	PT-discharged	3	0.9
	Death	1	0.3
	Attending	47	14.7
Review pattern	Not reviewed	250	77.9
	Once	58	18.1
	Twice	12	3.7
	Thrice	1	0.3
Outcome measure used	Yes	249	77.6
	No	72	22.4

Key:

PT: Physiotherapy

Women were found to have a significantly higher mean number of visits to their primary care clinic and diagnostic services than men.¹² Biological and hormonal differences contribute to this disparity. Post-menopausal hormonal changes predispose them to conditions such as osteoporosis, cardiovascular diseases, urogenital problems, and autoimmune disorders.¹³ These factors explain why women constitute a greater proportion of geriatric healthcare seekers, including those attending physiotherapy clinics.

The study also found that a high majority of the patients were able to provide their medical histories without assistance, indicating preserved cognitive and communicative abilities among most of them. This suggests that the majority did not suffer from neurological or cognitive impairments that would hinder self-reporting. According to Saher et al.,¹⁴ cognitive decline is common with aging, but since most patients in this study presented primarily with musculoskeletal conditions, their mental function remained largely intact. Nevertheless, a small proportion of case files lacked documentation of the source of history, highlighting the need for physiotherapists to adopt consistent documentation practices, as this improves the credibility of clinical data and continuity of care.

The analysis of the diagnoses revealed that Osteoarthritis (OA) was the most frequently managed condition, either as a single or co-existing diagnosis. OA has been identified as the leading cause of disability among older adults globally and in Nigeria.¹⁵ Several studies corroborate this trend.^{16,17} Physiotherapy is a cornerstone in OA management, primarily aimed at pain reduction, maintenance of joint mobility, and prevention of disability.^{18,19} The prevalence of OA increases with age and is particularly common in weight-bearing joints such as the knees, hips, and spine.²⁰ The present study found that knee OA was the most common

presentation, consistent with the literature, which identifies the knee as the most frequently affected site in older adults.²¹ The second most prevalent condition was lumbar spondylosis, which manifests mainly as low back pain.^{22,23} This degenerative spinal disorder is closely associated with the ageing process, although other risk factors such as obesity, prior back injury, prolonged sitting, repetitive lifting, and whole-body vibration (as seen in drivers) have been.^{20,23} Cervical spondylosis, a similar degenerative condition affecting the neck region, was also documented but to a lesser extent.

Among neurological disorders, cerebrovascular accident (CVA or stroke) was the most frequent. Stroke is well known to increase in prevalence with age, and the growing elderly population globally contributes to rising stroke incidence.^{24,25} Managing older stroke survivors is often complex due to frailty and co-morbidities.^{26,27} Parkinson's disease was also recorded, though in relatively few cases. Nigerian studies have shown lower Parkinson's prevalence compared to European populations.²⁸ The low prevalence of Parkinson's disease in Sub-Saharan African countries like Nigeria may stem from several issues, including the relatively youthful population, with less than 5% of the total population above 65 years; the fact that most cases of Parkinson's disease are not recognized partly because of the assumption that neurological diseases are part of normal aging; and the scarcity of epidemiological studies.²⁹ Another notable trend was the high occurrence of multiple diagnoses, reflecting the phenomenon of multi-morbidity in geriatric patients. As individuals age, they accumulate multiple chronic conditions due to a combination of genetic, environmental, and lifestyle factors.³⁰ Finally, the presence of numerous files with undocumented diagnoses underscores the urgent need to strengthen record-keeping practices among physiotherapists, especially in specialized

geriatric facilities like CTAGC, to ensure accurate data for research and clinical auditing. The yearly distribution of patient records showed variations across the five-year period. The lowest number of cases was recorded in 2018, primarily due to missing or incomplete case files. The poor record quality for that year suggests inadequate archiving or loss of patient data. It was therefore recommended that more attention be given to proper maintenance of patient files, especially in multi-clinic facilities serving elderly patients who often have multiple follow-ups. A decline in patient attendance in 2020 and 2021 was also noted, which can be attributed to the COVID-19 pandemic. Restrictions on movement, fear of infection, and hospital visitation limitations during the pandemic led to reduced outpatient clinic attendance globally, particularly among older adults who were the most vulnerable group. Across all years, female patients consistently outnumbered males, reaffirming the demographic pattern of higher female survival and health service utilization. In terms of diagnosis distribution, OA remained the most prevalent condition from 2019 to 2022, while lumbar spondylosis was more common in 2018. This trend demonstrates the enduring burden of OA among older adults and aligns with global projections that its prevalence will double by 2050 and triple by 2100.^{16,17}

A striking observation in this study was the time lag between referral and first presentation at the physiotherapy clinic. In more than one-quarter of the cases, this delay ranged between two and four months. Such delays can negatively affect treatment outcomes, especially in degenerative and progressive conditions. However, once patients presented, all of them received their first treatment on the same day, suggesting that the clinic may have a policy of initiating intervention immediately to promote compliance and reduce dropout rates. The reasons for delayed presentation may include financial constraints, transportation challenges, fatigue, and the burden of multiple clinic visits,

as older adults often attend different hospital units for other chronic conditions.³¹ Therefore, establishing patient-friendly scheduling systems or integrated geriatric care models could reduce missed appointments and improve adherence.

Regarding treatment modalities, exercise therapy emerged as the most frequently used intervention. This is consistent with physiotherapy principles emphasizing movement re-education, strengthening, and functional restoration in older adults.³² Exercise is known to reduce pain, improve range of motion, and enhance muscle power and balance. The treatment goals recorded by physiotherapists frequently included phrases such as “pain management,” “improve range of motion,” “improve muscle power,” “improve function,” “back care education,” “postural correction,” and “prevent further complications.” The frequent use of brief phrases in documentation may reflect time constraints or high patient volume, but it still provides insight into the therapeutic focus of physiotherapy for geriatric populations.

The discharge process was found to be inconsistent, with many cases lacking documentation of discharge type. This gap may arise from self-discharge, where patients discontinue attendance without formal evaluation. Establishing a standardized discharge protocol would enhance documentation and continuity of care. Similarly, incorporating scheduled patient reviews after defined numbers of therapy sessions would facilitate better tracking of progress and outcomes.³³ The study also noted that the use of standardized outcome measures was a common practice at CTAGC. In evidence-based physiotherapy, the use of validated tools such as the Timed Up-and-Go (TUG) test, Sit-to-Stand test, and 6-Minute Walk Test is essential for objective assessment of functional progress.^{34,35} More efforts are required to widen the utilization of standardized outcome measures in the management of older persons, as the absence of

such measures will limit the ability to evaluate treatment effectiveness systematically. Therefore, the development and enforcement of a policy mandating the use of culturally appropriate outcome measures for geriatric patients are highly recommended.

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